

2	(a) (i) 1. distance of path / along line AB	B1 [1]
	2. shortest distance between AB / distance in straight line between AB or displacement from A to B	B1 [1]
	(ii) acceleration = rate of change of velocity	A1 [1]
(b) (i)	distance = area under line or $(v/2)t$ or $s = (8.8)^2 / (2 \times 9.81)$ = $8.8 / 2 \times 0.90 = 3.96 \text{ m}$ or $s = 3.95 \text{ m} = 4(.0) \text{ m}$	C1 A1 [2]
	(ii) acceleration = $(-4.4 - 8.8) / 0.50$ = $(-) 26(.4) \text{ m s}^{-2}$	C1 A1 [2]
(c) (i)	the accelerations are constant as straight lines	B1
	the accelerations are the same as same gradient or no air resistance as acceleration is constant or change of speed in opposite directions (one speeds up one slows down)	B1 [2]
	(ii) area under the lines represents height or KE at trampoline equals PE at maximum height	B1
	second area is smaller / velocity after rebound smaller hence KE less	B1
	hence less height means loss in potential energy	A0 [2]